

Linear Algebra — Homework 3

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Section A

Solutions:

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Part 1

Q 1:

Linear Equation of each Node:

$$500 = x_1 + x_2 \quad \dots \quad \textcircled{A}$$

$$x_1 + 200 = x_3 + x_4 \quad \dots \quad \textcircled{B}$$

$$x_2 + x_3 = 400 + x_5 \quad \dots \quad \textcircled{C}$$

$$x_4 + x_5 = 300 \quad \dots \quad \textcircled{D}$$

Q 2:

Setting up Augmented Matrix

$$x_1 + x_2 = 500$$

$$x_1 - x_3 - x_4 = -200$$

$$x_2 + x_3 - x_5 = 400$$

$$x_4 + x_5 = 300$$

$$\therefore A \cdot x = b$$
$$\begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & -1 & -1 & 0 \\ 0 & 1 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 500 \\ -200 \\ 400 \\ 300 \end{bmatrix}$$

$$\left[\begin{array}{cccccc|c} 1 & 1 & 0 & 0 & 0 & 500 \\ 1 & 0 & -1 & -1 & 0 & -200 \\ 0 & 1 & 1 & 0 & -1 & 400 \\ 0 & 0 & 0 & 1 & 1 & 300 \end{array} \right]$$

Question 3.

General Solution & Parametric Form

$$R_2' = R_2 - R_1$$

$$\left[\begin{array}{cccccc|c} 1 & 1 & 0 & 0 & 0 & 500 \\ 0 & -1 & -1 & -1 & 0 & -700 \\ 0 & 1 & 1 & 0 & -1 & 400 \\ 0 & 0 & 0 & 1 & 1 & 300 \end{array} \right]$$

$$R_3' = R_2 + R_3$$

$$\left[\begin{array}{cccccc|c} 1 & 1 & 0 & 0 & 0 & 500 \\ 0 & -1 & -1 & -1 & 0 & -700 \\ 0 & 0 & 0 & -1 & -1 & -300 \\ 0 & 0 & 0 & 1 & 1 & 300 \end{array} \right]$$

2.

$$R_4' = R_3 + R_4$$

$$\left[\begin{array}{cccccc|c} 1 & 1 & 0 & 0 & 0 & 500 \\ 0 & -1 & -1 & -1 & 0 & -700 \\ 0 & 0 & 0 & -1 & -1 & -300 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

$$R_2' = -R_2 \quad \& \quad R_3' = -R_3$$

$$\left[\begin{array}{cccccc|c} 1 & 1 & 0 & 0 & 0 & 500 \\ 0 & 1 & 1 & 1 & 0 & 700 \\ 0 & 0 & 0 & 1 & 1 & 300 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

$$R_1' = R_2 - R_4$$

$$\left[\begin{array}{cccccc|c} 1 & 0 & 1 & 1 & 0 & 200 \\ 0 & 1 & 1 & 1 & 0 & 700 \\ 0 & 0 & 0 & 1 & 1 & 300 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

$$R_1' = R_3 - R_1 \quad \& \quad R_2' = R_2 - R_3$$

$$\left[\begin{array}{cccccc|c} 1 & 0 & -1 & 0 & -1 & 100 \\ 0 & 1 & 1 & 0 & -1 & 400 \\ 0 & 0 & 0 & 1 & 1 & 300 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

$$x_1 - x_3 + x_5 = 100 \quad \Rightarrow \quad x_1 = 100 + x_3 - x_5$$

$$x_2 + x_3 - x_5 = 400 \quad \Rightarrow \quad x_2 = 400 - x_3 + x_5$$

$$x_4 + x_5 = 300 \quad \Rightarrow \quad x_4 = 300 - x_5$$

$$x_5 = x_5 ; x_3 = x_3$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 100 + x_3 - x_5 \\ 400 - x_3 + x_5 \\ x_3 \\ 300 - x_5 \\ x_5 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 100 \\ 400 \\ 0 \\ 300 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 1 \\ -1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} -1 \\ 1 \\ 0 \\ -1 \\ 1 \end{bmatrix}$$

$\vec{b}$ $\vec{v}$ $\vec{u}$

Q4: General Solution:-

$$x = x_5 \vec{u} + x_3 \vec{v} + \vec{b}$$

where $\vec{u}$ & $\vec{v}$ are SD vectors and $\vec{b}$ is constant

data. Here x_3 & x_5 are scalars $\in \mathbb{R}$

• ' $\vec{b}$ ' represents valid routing configuration that satisfies every nodes conservation law.

• $\vec{u}$ & $\vec{v}$ represents the internal data flow.

• x_3 & x_5 are free variables $\in \mathbb{R}$.

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Part 2

Audio Signal Compression

Q1.

$$S\vec{x} = \vec{0}$$

$$s_1 = \begin{bmatrix} 1 \\ 2 \\ 1 \\ 1 \end{bmatrix}, \quad s_2 = \begin{bmatrix} 2 \\ 4 \\ 0 \\ -2 \end{bmatrix}, \quad s_3 = \begin{bmatrix} 4 \\ 8 \\ 2 \\ 0 \end{bmatrix}, \quad s_4 = \begin{bmatrix} 0 \\ 0 \\ 2 \\ 4 \end{bmatrix}$$

$$S = [s_1, s_2, s_3, s_4]$$

$$S = \begin{bmatrix} 1 & 2 & 4 & 0 \\ 2 & 4 & 8 & 0 \\ 1 & 0 & 2 & 2 \\ 1 & -2 & 0 & 4 \end{bmatrix}$$

• Augmented Matrix:

$$\left[\begin{array}{cccc|c} 1 & 2 & 4 & 0 & 0 \\ 2 & 4 & 8 & 0 & 0 \\ 1 & 0 & 2 & 2 & 0 \\ 1 & -2 & 0 & 4 & 0 \end{array} \right]$$

$$R_2' = 2R_1 - R_2 ; R_3' = R_1 - R_3 , R_4' = R_1 - R_4$$

$$\begin{bmatrix} 1 & 2 & 4 & 0 & : & 0 \\ 0 & 0 & 0 & 0 & : & 0 \\ 0 & 2 & 2 & -2 & : & 0 \\ 0 & 4 & 4 & -4 & : & 0 \end{bmatrix}$$

$$R_4' = 2R_3 - R_4$$

$$\begin{bmatrix} 1 & 2 & 4 & 0 & : & 0 \\ 0 & 0 & 0 & 0 & : & 0 \\ 0 & 2 & 2 & -2 & : & 0 \\ 0 & 0 & 0 & 0 & : & 0 \end{bmatrix}$$

Put zero rows at end & $R_3' = 1/2 R_3$

$$\begin{bmatrix} 1 & 2 & 4 & 0 & : & 0 \\ 0 & 1 & 1 & -1 & : & 0 \\ 0 & 0 & 0 & 0 & : & 0 \\ 0 & 0 & 0 & 0 & : & 0 \end{bmatrix}$$

$$R_1' = R_1 - 2R_2$$

$$\begin{bmatrix} 1 & 0 & 2 & 2 & : & 0 \\ 0 & 1 & 1 & -1 & : & 0 \\ 0 & 0 & 0 & 0 & : & 0 \\ 0 & 0 & 0 & 0 & : & 0 \end{bmatrix}$$

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2 pivot columns out of 4.

4.

Q2. Prove Linear Independence

An indexed set of vectors in $\mathbb{R}^n$ is said to be linearly independent if the vector equation only has the trivial solution (one unique solution).

- However, in our ^{system} set of linear equations, we've s_3 & s_4 as free variables. We got only 2 pivot columns out of 4 columns. A homogeneous equations with free variables have always non-trivial solutions.
- Hence $s_1 = 0$ is linearly dependent.

Q3.

$$s_1 + 2s_3 + 2s_4 = 0 \Rightarrow s_1 = -2(s_3 + s_4)$$

$$s_2 + s_3 + s_4 = 0 \Rightarrow s_2 = -(s_3 + s_4)$$

Q4.

The pivot columns of a matrix form a basis for its column space. The pivot columns were s_3 & s_4 , hence these are the basis signals $\{s_3, s_4\}$.

- To transmit only independent signals, we will use only s_3 & s_4 .

$$s_2 = -(s_3 + s_4)$$

$$-s_2 - s_4 = s_3$$

$$\Rightarrow s_3 = -(s_2 + s_4)$$

$$\text{Put } s_3 = -(s_2 + s_4) \text{ in } s_1 = -2(s_3 + s_4)$$

$$s_1 = -2(-s_2 + s_4) + s_4$$

$$s_1 = 2s_2 + 2s_4 + s_4$$

$$s_1 - 2s_2 = 3s_4$$

$$s_4 = \frac{s_1 - 2s_2}{3}$$

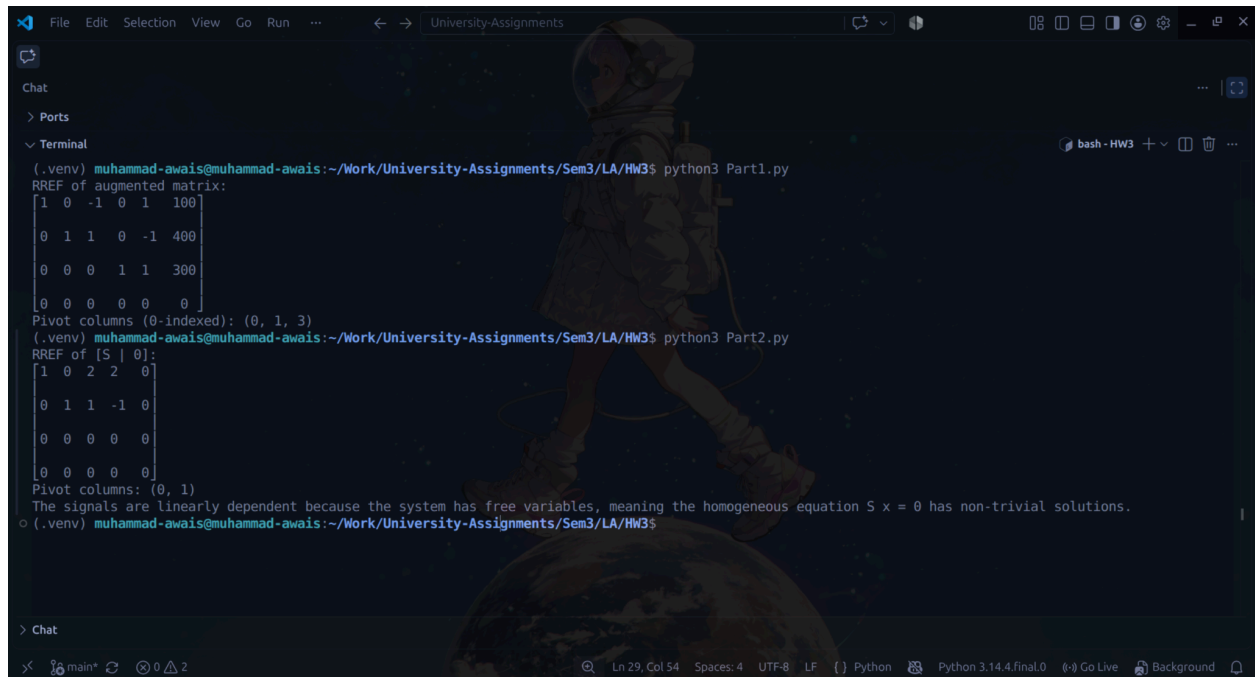
$$\text{Put } s_4 = \frac{s_1 - 2s_2}{3} \text{ in } s_3 = -(s_2 + s_4)$$

$$s_3 = -\left(s_2 + \frac{s_1 - 2s_2}{3}\right)$$

$$s_3 = -\left(\frac{3s_2 - s_1 + 2s_2}{3}\right)$$

$$s_3 = -\left(\frac{5s_2 - s_1}{3}\right)$$

Python Verification:



The screenshot shows a VS Code editor window titled "University-Assignments". The terminal window is active, showing the following output:

```
(.venv) muhammad-awais@muhammad-awais:~/Work/University-Assignments/Sem3/LA/HW3$ python3 Part1.py
RREF of augmented matrix:
[ 1  0 -1  0  1 100]
[ 0  1  1  0 -1 400]
[ 0  0  0  1  1 300]
[ 0  0  0  0  0  0]
Pivot columns (0-indexed): (0, 1, 3)
(.venv) muhammad-awais@muhammad-awais:~/Work/University-Assignments/Sem3/LA/HW3$ python3 Part2.py
RREF of [S | 0]:
[ 1  0  2  2  0]
[ 0  1  1 -1  0]
[ 0  0  0  0  0]
[ 0  0  0  0  0]
Pivot columns: (0, 1)
The signals are linearly dependent because the system has free variables, meaning the homogeneous equation  $Sx = 0$  has non-trivial solutions.
(.venv) muhammad-awais@muhammad-awais:~/Work/University-Assignments/Sem3/LA/HW3$
```

The terminal window also shows a chat panel on the left and a status bar at the bottom indicating the current file is "main*" and the editor is in "Python" mode.